

LANNION COTE DE GRANIT, France

WMO: 071180

Lat: 48.7553N

Lon: 3.4686W

Elev: 290

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.7	32.0	23.9	18.3	33.6	26.5	20.7	35.3	47.3	51.8	41.5	50.5	6.3	150	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	76.4	65.4	72.8	63.9	70.1	62.7	67.0	73.5	65.3	70.7	63.9	68.4	9.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.2	69.2	63.1	87.5	67.5	62.1	84.3	66.3	31.6	73.8	30.3	70.6	29.3	68.7	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.9	23.7	20.8	DB	25.1	84.3	3.0	5.3	23.0	88.1	21.2	91.2	19.6	94.2	17.4	98.1
			WB	24.3	71.0	3.3	2.8	22.0	73.0	20.0	74.6	18.2	76.2	15.8	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	44.3	44.0	46.4	49.4	53.8	58.6	61.9	61.6	59.3	55.0	49.2	45.3
	DBStd	8.01	5.30	5.08	4.85	4.58	4.37	4.25	3.73	3.76	4.52	4.90	4.63	5.47
	HDD50	815	184	176	131	64	13	1	0	0	2	13	69	162
	HDD65	4621	641	589	576	468	347	196	111	119	178	310	475	611
	CDD50	1709	8	7	20	45	131	260	368	359	282	169	44	16
	CDD65	42	0	0	0	0	1	5	14	13	8	1	0	0
	CDH74	250	0	0	0	3	13	47	80	64	36	7	0	0
	CDH80	53	0	0	0	0	2	10	20	13	6	2	0	0
Wind	WSAvg	9.8	11.6	11.2	10.6	9.5	9.4	8.7	8.8	8.5	8.2	9.1	10.5	11.1
Precipitation	PrecAvg	33.9	3.7	3.3	2.2	2.6	2.3	2.0	2.0	2.2	1.8	3.5	4.1	4.3
	PrecMax	44.0	6.7	6.4	6.4	6.8	4.5	4.6	4.6	5.8	5.3	7.5	8.0	9.3
	PrecMin	24.2	0.8	0.6	0.3	0.8	0.6	0.3	0.8	0.7	0.2	0.6	1.1	0.9
	PrecStd	4.5	1.7	1.6	1.3	1.5	1.0	1.1	1.0	1.1	1.2	1.6	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	60.2	65.1	70.9	75.1	80.2	82.2	80.8	79.0	71.8	62.7	58.6
		MCWB	53.8	52.3	54.6	58.9	64.1	67.0	67.9	67.2	65.9	64.0	58.9	54.7
	2%	DB	55.1	55.5	59.4	64.7	68.0	73.2	75.1	73.8	73.0	66.8	60.1	56.8
		MCWB	53.0	51.4	52.6	55.6	59.5	63.8	65.1	65.1	64.0	61.1	56.8	54.2
	5%	DB	53.7	53.3	56.0	60.3	64.5	69.2	71.7	70.4	69.8	64.2	57.6	55.2
		MCWB	51.7	50.5	51.7	53.6	57.8	61.8	63.5	63.8	62.8	59.9	55.0	53.2
	10%	DB	52.1	51.6	53.8	57.0	61.3	66.3	69.2	68.1	66.9	62.2	56.1	53.5
		MCWB	50.1	49.2	50.5	51.8	56.1	60.4	62.4	62.2	61.3	58.7	53.6	51.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	54.3	56.0	59.5	64.5	68.9	68.9	69.7	67.5	64.5	60.1	56.2
		MCDB	55.7	57.6	61.4	67.5	73.3	78.3	78.4	79.1	73.8	68.7	61.8	57.4
	2%	WB	53.6	52.4	54.0	56.4	60.3	64.9	66.3	66.8	65.4	62.4	57.3	54.8
		MCDB	54.9	54.6	57.6	63.2	65.8	70.7	73.0	72.2	70.8	65.9	59.3	56.1
	5%	WB	52.1	51.0	52.5	54.4	58.5	62.8	64.7	64.9	63.8	60.7	55.6	53.5
		MCDB	53.4	53.0	55.6	59.1	63.3	68.1	70.1	68.9	68.3	63.4	57.3	55.0
	10%	WB	50.5	49.5	50.9	52.6	56.8	60.9	63.3	63.4	62.2	59.3	54.0	51.9
		MCDB	51.8	51.4	53.6	56.4	60.7	65.1	67.9	66.9	65.9	61.7	55.8	53.6
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.9	11.5	13.5	13.2	13.3	13.1	13.3	14.1	12.1	10.1	9.1
		MCDBR	9.5	12.8	15.6	19.1	18.3	18.6	18.2	17.4	18.2	14.1	11.7	9.5
		MCWBR	9.1	9.6	10.4	11.6	10.9	11.0	9.9	9.6	10.0	9.1	9.3	8.6
	5% WB	MCDBR	9.0	11.0	13.9	17.0	16.4	17.1	16.1	15.2	15.9	12.9	10.9	9.1
MCWBR		9.1	9.3	9.9	11.0	10.4	10.8	9.8	9.3	9.7	9.0	9.5	9.0	
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.374	0.384	0.388	0.382	0.381	0.371	0.360	0.354	0.333	0.324	
	taud	2.413	2.393	2.326	2.316	2.352	2.395	2.400	2.449	2.460	2.467	2.444	2.378	
	Ebn at Noon	230	254	265	274	277	279	277	276	267	248	227	214	
	Edh at Noon	21	27	34	37	37	36	36	32	29	25	20	19	
All-Sky Solar Radiation	RadAvg	298	544	895	1395	1662	1823	1744	1475	1158	667	366	250	
	RadStd	25	63	93	131	132	196	103	118	84	53	36	30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page